REFERENCE FILE

# REPORT DOCUMENTATION PAGE

Form Approved
OMB No. 0704-0188

Public reporting burden for this collection of information is estimated to average 1 hour per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503.

| 1. AGENCY USE ONLY (Leave blank) | 2. REPORT DATE | 3. REPORT TYPE AND DATES COVERED |
|---|---|---|
| | 6 May 93 | Final |

**4. TITLE AND SUBTITLE**
Test Operations Procedure (TOP)
6-2-504 Maintainability (Communications/Electronics)

**5. FUNDING NUMBERS**
WU A 268445

**6. AUTHOR(S)**

**7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES)**
Commander
U.S. Army Electronic Proving Ground
ATTN: STEEP-MO
Fort Huachuca, AZ 85613-7110

**8. PERFORMING ORGANIZATION REPORT NUMBER**
Test Operations Procedure (TOP) 6-2-504

**9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES)**
Commander
U.S. Army Test and Evaluation Command
ATTN: AMSTE-CT-T
Aberdeen Proving Ground, MD 21005-5055

**10. SPONSORING/MONITORING AGENCY REPORT NUMBER**
Same as item 8.

**11. SUPPLEMENTARY NOTES**

**12a. DISTRIBUTION/AVAILABILITY STATEMENT**
Approved for public release;
distribution unlimited.

**12b. DISTRIBUTION CODE**

**13. ABSTRACT** *(Maximum 200 words)*

This TOP presents a general methodology to monitor, review and verify quantitative and qualitative maintainability characteristics of electronic test items. These test procedures apply to all maintainability/supportability qualification tests and demonstrations which are conducted/witnessed by TECOM Test Centers. These include both government and contractor testing in support of research, development and acquisition programs.

**14. SUBJECT TERMS**
Communication, electronics, maintainability, physical teardown/logistic demonstrations

**15. NUMBER OF PAGES**
32

**16. PRICE CODE**

| 17. SECURITY CLASSIFICATION OF REPORT | 18. SECURITY CLASSIFICATION OF THIS PAGE | 19. SECURITY CLASSIFICATION OF ABSTRACT | 20. LIMITATION OF ABSTRACT |
|---|---|---|---|
| UNCLASSIFIED | | | SAR |



REFERENCE FILE

Source: https://assist.dla.mil -- Downloaded: 2025-01-21T14:51Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE (TOP)

*Test Operations Procedure (TOP) 6-2-504 6 August 1993
AD No. A273137

# MAINTAINABILITY (COMMUNICATIONS/ELECTRONICS)


| | | | Page |
|---|---|---|---|
| Paragraph | 1. | SCOPE | 1 |
| | 2. | FACILITIES AND INSTRUMENTATION | 2 |
| | 2.1 | Facilities | 2 |
| | 2.2 | Instrumentation | 2 |
| | 3. | REQUIRED TEST CONDITIONS | 2 |
| | 3.1 | Maintainability Demonstration (M-Demo) | 2 |
| | 3.2 | Physical Teardown/Logistic Demonstration (PT/LD) | 2 |
| | 3.3 | Required Resources | 2 |
| | 3.4 | Preparation For Test | 3 |
| | 4. | PROCEDURES | 6 |
| | 4.1 | Initial Inspection | 6 |
| | 4.2 | Maintainability Demonstration (M-Demo) | 6 |
| | 4.3 | Physical Teardown/Logistic Demonstration (PT/LD) | 9 |
| | 5. | DATA REQUIRED | 11 |
| | 5.1 | Maintainability Demonstration (M-Demo) | 11 |
| | 5.2 | Physical Teardown/Logistic Demonstration (PT/LD) | 11 |
| | 6. | PRESENTATION OF DATA | 12 |
| | 6.1 | General | 12 |
| | 6.2 | Maintainability Demonstration (M-Demo) | 12 |
| | 6.3 | Physical Teardown/Logistic Demonstration (PT/LD) | 13 |
| | 6.4 | Data Forms | 15 |
| APPENDIX | A. | BACKGROUND | A-1 |
| | B. | DATA COLLECTION FORMS | B-1 |
| | C. | QUALITY INDEX RULES AND CRITERIA | C-1 |
| | D. | REFERENCES | D-1 |


## 1. SCOPE

a. This test operations procedure presents a general methodology to monitor, review, and verify quantitative and qualitative maintainability characteristics of electronic test items. These test procedures apply to all maintainability/supportability qualification tests and demonstrations which are conducted/witnessed by TECOM Test Centers. These include both government and contractor testing in support of research, development, and acquisition programs.

b. Limitations. Since unique contractual obligations apply to each acquisition program some of the guidelines and methods contained in this test operations procedure may not be invoked without legal implication. Test personnel must familiarize themselves with the general contract requirements when

*This TOP supersedes MTP 6-2-504, March 1970
Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

monitoring and witnessing contractor demonstrations. The program office representative will be the designated point of contact for questions concerning the delivery requirements and scope of work.

## 2. FACILITIES AND INSTRUMENTATION.

### 2.1 Facilities.

Test facilities must be representative of the proposed field conditions for operation and maintenance of the test item/system. The correlation between maintainability characteristics and measured values is uniquely related to the environment in which the tasks are performed. Typically, unit level maintenance tasks are performed under field conditions, and direct support (DS) tasks are performed in a representative maintenance facility.

### 2.2 Instrumentation.

No unique instrumentation is required to conduct the procedures outlined in this test operations procedure. Each test item will have independent instrumentation requirements as directed by the maintenance publications. These must be available to perform the required maintenance procedures.

## 3. REQUIRED TEST CONDITIONS.

These test procedures apply to all maintainability/supportability qualification tests and demonstrations which are conducted/monitored/witnessed by TECOM Test Centers. These include both government and contractor testing in support of research, development, and acquisition programs.

### 3.1 Maintainability Demonstration (M-Demo).

This demonstration provides certification that developed items meet the maintainability design specifications. Technically, this is accomplished through component fault insertion and statistical analysis of appropriate (usually organizational and direct support) maintenance levels.

### 3.2 Physical Teardown/Logistic Demonstration (PT/LD).

This demonstration is used to review system supportability through analysis of the maintainability and support considerations. Technically, this is accomplished through insertion of non-destructive faults for verification of troubleshooting procedures, physical disassembly and reassembly of the system elements, and comparative analysis.

### 3.3 Required Resources.

a. Test item and associated support equipment configured to represent the end item system under proposed operation/maintenance conditions. Specific



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

requirements are defined by MIL-STD-471A[1], page 6, para 3-28.

b. System Support Package (SSP). This is the composite of all required draft equipment publications; technical manuals; repair parts; special and common tools; tool kits; support and handling equipment; test measurement and diagnostic equipment; shop facilities; training; and personnel requirements as defined by AR-700-127[2], page 18, paras 3-31 and 3-32.

c. Military operators, maintainers or government personnel with the required skills and training as identified by MIL-STD-471A[1], page 7, para 4.1.2.9.

d. Stopwatch (digital or analog) and clock with second hand for timing maintenance tasks and overall task time within minutes and seconds.

e. Automated data processing equipment and application software for processing of Test Incident Reports (TIRs), tracking of maintainability data, correlation of test anomalies, and assignment of corrective actions.

f. Project Notebook for historical record of test events.

g. Photographic support is not required under normal test conditions. However, a 35mm camera, electronic flash, and zoom lens with macro capability may be helpful in communicating test conditions or reporting maintenance/ logistics anomalies that are detected.

3.4 Preparation For Test.

The test conductor must assure that the following essential planning issues are addressed. TECOM test center representatives will verify that all of these items are completed and advise the Program Manager (PM) of any shortcomings. All actions should be completed no later than 30 days prior to test.

a. The master test schedule must provide adequate time for completion of testing, analysis of test data, and initiation of corrective action prior to developmental test (DT). Completion of the M-Demo and PT/LD 6 months prior to DT is a standard milestone which will allow a minimum of 4 months for resolution of test issues. Uncorrected maintenance deficiencies and associated logistic problems are detrimental to the program since they impact the Reliability, Availability and Maintainability (RAM) values which become part of the formal test record.

b. A safety review must be completed 30 days prior to the start of any testing and a safety release issued to the responsible test activity to assure the safety of TECOM civilian and Soldier-Operator-Maintainer Test and Evaluation (SOMTE) personnel.

[1]Reference numbers match those in Appendix D, References.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

c. Appropriate environmental documentation (IAW AR 200-2[3], Environmental Effects of Army Actions) must be prepared for any field testing and an environ-mental release issued to the responsible test activity prior to start testing.

d. A contractor-developed test plan must be coordinated and approved at least 30 days prior to test. Government acceptance of the test plan should be annotated by concurrence letter.

(1) The responsible test center must coordinate with the PM to ensure that a copy of the test plan is provided for review and comment prior to the start of test.

(2) The technical content of the test plan must conform to the appropriate regulations and standards for the type of test. Army Regulation 750-1[4] defines maintenance policy and maintenance support requirements. MIL-STD-721C[5] defines maintainability terms.

e. A candidate fault list must be developed and included as an appendix to the test plan. MIL-STD-471A[1] provides fault stratification guidelines.

(1) When component fault insertion is required, this list is devel-oped from reliability predictions and failure mode analysis. This stratified list provides a proportional sample of maintenance tasks based on probability of failure. Simple designs which contain few replaceable units will probably result in multiple remove and replace actions involving the same module. Since the measured replacement time for these modtules should be relatively constant, the most significant element becomes the diagnostic segment of the maintenance action. The stratified list of candidate faults must adequately consider all failure modes regardless of the resultant corrective action.

(2) The same guidelines apply to non-destructive faults except when the key issue is validation of troubleshooting procedures. The candidate fault list must also consider the number of processes and flow charts to be reviewed and provide a sample of tasks which proportionally represents the type of failure weighted against the probability of failure.

(3) When fault insertion is not required the candidate list should be developed from the Logistic Support Analysis Record (LSAR) and the Maintenance Allocation Chart (MAC). Particular consideration should be given to the items with the highest remove and replace probability, longest maintenance times, and most complex maintenance procedures. MIL-STD-1388-1[6] and MIL-STD-1388-2[7] provide LSA/LSAR requirements.

f. The test witness, as a significant player in the certification process and a subject matter expert, is responsible for advising the PM of any inadequacies in the test plan and procedures. TECOM Test Centers must ensure compliance with test standards and avoid witnessing of substandard tests in view of possible legal implications. Test centers must notify HQ TECOM of



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

inadequate test plans/procedures whenever local resolution efforts fail so that command level influence can be brought to bear. In the past, government witness signatures on data sheets have been construed to be government acceptance of the tests. This forced PMs to fund additional development effort to correct problems thus paying twice for qualification testing instead of receiving contractual consideration for a deficient product.

g. The test team must be developed and personnel requirements identified to assure that appropriate military personnel are available to support the test. Utilization of contractor personnel to perform operation and maintenance tasks which are normally done by military personnel is not a valid approach for certifying maintainability values. Military personnel may be provided by the responsible test center if the test support requirements and schedule are identified early. This represents the best value to the government since personnel that are routinely trained prior to developmental test (DT) are typically available to the PM prior to the start of DT.

h. Operator and Maintainer Training must be completed and adequate familiarization time allocated.

i. Adequate contractor or government facilities and equipment to support the testing must be identified and allocated. The test environment must represent the user configuration. Therefore, coordination is necessary for shop facilities, tools, test equipment, spares, power sources, COMSEC items, vehicles, aircraft, and other items as needed to support the representative configuration. Additionally, administrative facilities and support items for test conduct will be required.

j. Equipment technical manuals must be provided. Verified publications provide the highest confidence level in the operation and maintenance instructions. However, redline publications are acceptable provided they are validated and accepted during the contractural 90 percent in process review (IPR). Technical manuals which do not meet this requirement are normally too rudimentary to determine maintainability.

k. A proposed test schedule is required to adequately plan support requirements and manage the test activities. A good rule of thumb is to allocate one day of testing for each five maintenance tasks. The test duration is dependant upon the minimum sample size and statistical test method identified. Two administrative days are required in the schedule. One day should be allocated for set up, configuration inspection, diagnostic certification of the system, and a pretest briefing. The second day should be allocated for diagnostic certification of the system, teardown and packing, review of test observations, and the post test briefing.

l. Prepare the required test logs, maintenance data sheets, area of concern data sheets, and acquire any automation equipment or software required to support test administration and report writing.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

m. Identify and coordinate any special security requirements.

n. Coordinate photographic coverage if appropriate.

4. PROCEDURES.

4.1 Initial Inspection.

The responsible test activity and test center representative(s) will complete the following actions prior to starting the test:

a. Conduct an initial inspection of the test item and record the serial numbers of major line replaceable units (LRUs), software version numbers, and associated support equipment. Form B-1, Appendix B.

b. Set up the test item in a representative configuration using the instructions in the system technical manuals. Deviations to the setup instructions must be documented in the Project Notebook and a justification should be provided since this may impact the resultant maintainability values. Form B-2, Appendix B.

c. Run any internal built-in-test (BIT) routines and applicable diagnostic programs and document the test results. The normal operating and diagnostic procedures available to the crew and maintenance personnel should be used to certify the system's operational state before any faults are inserted. Form B-3, Appendix B.

d. Record data pertaining to Test Team personnel. Form B-4, Appendix B.

e. Inventory the SSP and record any shortages in the Project Notebook.

4.2 Maintainability Demonstration (M-Demo).

a. This test is conducted under a strict set of rules provided in MIL-STD-471A[1]. These guidelines assure statistically valid test results by providing a controlled mechanism for data acquisition. Any deviation from the standard is considered a significant issue since it typically causes the consumer to accept a greater risk that the item will not meet the user's needs.

b. A pretest briefing to all attendees will be conducted prior to the start of testing. This provides an opportunity to review the test procedures, introduce the Test Team, complete an attendance roster, discuss the ground rules and administrative details, and present the test schedule. This meeting also serves as a readiness review. Therefore, this is the appropriate forum to identify and resolve any technical issues or problems. Topics of discussion should include SSP shortages and potential test conduct errors since these are significant issues which must be resolved prior to starting the test.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

c. A government Test Director provides the overall management of the test and guides the test activities. This individual normally represents the PM and has the cognizant authority to provide government guidance to contractor personnel.

d. Maintenance tasks must be selected from the candidate fault list immediately prior to the demonstration. Each task is assigned a formal task number which will remain with it throughout the demonstration. Care must be taken to avoid compromising the test results by uncontrolled distribution of the selected tasks. Since adequate time to coordinate the engineering and technical support for fault insertion is needed, a reasonable time limit for distribution of the maintenance task list should be agreed upon by the Test Director and the Test Conductor. Releasing the list 2 weeks prior to the demonstration should be adequate if the candidate faults are properly validated.

e. The test center representative is responsible for witnessing the test, determining the statistical validity of the test procedures, and advising the Test Director of any discrepancies noted. Procedural issues will be discussed with the Test Conductor and corrected through mutual agreement and guidance from the Test Director.

f. Witnesses must actively participate in the data authentication process by keeping times, historical notes, and recording test observations. Official times are the responsibility of the designated Timekeeper/Data Recorder. The test witness times are utilized as a tool to verify the time recorded by the Test Team. Any timekeeping discrepancies should be discussed with the Test Director. Test observations should provide sufficient detail to allow the preparation of TIRs.

g. Operator and maintenance personnel should be debriefed by the Test Conductor at the conclusion of each maintenance task where problems are apparent. This will reduce confusion and allow for more accurate data collection since it becomes more difficult to distinguish the individual tasks as the volume of maintenance increases. Significant discussion of the maintenance task must be deferred until the test personnel are sequestered out of sight and hearing of the workplace. This will prevent compromising the fault in the event a retest is required.

h. Retests of demonstration tasks will retain the same unique number assigned during the selection process but will be annotated as "Retest." A typical example of task identification is "M-Demo Task 5", which can become "M-Demo Task 5a (Retest)". This prevents the same maintenance task from being selected more than once if additional tasks are selected from the pool of candidate faults. The number of retests should be held to a minimum by assuring the task is adequately prepared prior to demonstration. Tasks which result in damage to the test item will not be eligible for retest until an engineering review is completed and the fault is recertified. Multiple retests of the same task indicate the maintainability characteristics of the task are still



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

in a development status.

i. A post test briefing will be conducted at the conclusion of test activities. The test observations should be reviewed and appropriate comments solicited from test observers. Preliminary test findings may be presented if they are available .

j. The following "test conduct" errors occur frequently. Since these errors detract from the data credibility, "on the spot" corrections should be made when they are identified.

(1) Control the test environment by limiting discussion and assistance to the Test Team members. The Test Conductor provides the operator and maintainer interface during the maintenance tasks. All outside discussion is through the Test Conductor.

(2) Maintain adequate workspace by limiting the number of observers in the immediate area. Test personnel are entitled to a work environment which allows them free access to the equipment without distraction and outside influence.

(3) Assure all operators and maintainers are sequestered when faults are being inserted and discussed. Allowing for a second operator and maintainer provides an unbiased sample when maintenance tasks are retested.

(4) Time-out procedures must be defined and rules established to determine who can call a "time out" and under what circumstances they are appropriate. The operator or maintainer must understand the procedures and

retain the right to advise the Test Conductor when a time out is needed. The test environment must allow the Timekeeper to record the time accurately.

(5) Discussion, during the maintenance task, is limited to interpretation of the technical instructions and maintenance procedures. Care must be taken to avoid compromising the test results by providing maintenance guidance and troubleshooting procedures.

(6) Technical manual errors, omissions, and ambiguous instructions may be corrected with redline changes during the demonstration. However, if discussion is required to determine the correct procedure, the task in progress must be aborted. It can be retested later in the demonstration using the same maintainer or immediately retested if a second maintainer is available.

(7) Retests must retain the same task number as the original. The chronological sequence of tasks can vary but it should be evident through the numbering system how many tasks are performed. This practice will ensure that the minimum sample of tasks required by the analytical method is achieved. More maintenance actions may occur than number of actual tasks performed if



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

retests take place, tasks are thrown out, or additional tasks are selected.

(8) Spares must be provided to effect the interchange of components required by the troubleshooting procedures. Recorded times must include the additional maintenance time to reinstall the original component, if the maintenance action results in an ambiguity greater than one, and perform additional fault isolation. Maintenance times must be actually demonstrated rather than being constructed via piecemeal or fragmented data from varied sites and times.

(9) A representative system configuration must be provided for all "on system" maintenance tasks. There is no correlation between maintenance performed on a workbench and maintenance performed on the system. A scale mock up can be utilized if system hardware is not available.

4.3 Physical Teardown/Logistic Demonstration (PT/LD).

a. This test is not conducted under a distinct set of rules defined by MIL-STD. Historically, there is little consistency in the test conduct, technical content, or analytical procedures used during the PT/LD since each program determines their unique test requirements. The general rules of conduct provided in MIL-STD-471A[1] can be applied to the maintainability assessment portion of this demonstration in a tailored interactive approach since several data issues and subject disciplines are addressed simultaneously.

b. A government Test Director provides the overall management of the test and guides the test activities. This individual normally represents the PM and has the cognizant authority to provide government guidance to contractor personnel.

c. A pretest meeting will be conducted prior to the start of testing. This provides an opportunity to review the test procedures, introduce the Test Team, complete an attendance roster, discuss the ground rules, administrative details, and present the test schedule. This meeting also serves as a readiness review. Therefore, this is the appropriate forum to identify and resolve any technical issues or problems.

d. Measurement of maintainability characteristics is the primary issue during physical teardown. The measured data should be reviewed and incorporated into the LSAR in place of predicted data to validate and further refine the maintenance concept and support structure. Ideally, every remove and replace task is a prospective candidate for demonstration and measurement. If

adequate spares are available all LRU/SRUs can be interchanged with like spares providing both maintainability and interchangeability data.

e. Maintenance tasks must be selected from the candidate fault list immediately prior to the demonstration. It is not essential to assign a formal task number at this time since the sequence of tasks may be negotiated



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

to optimize the test schedule. Care must be taken to avoid compromising the test results by uncontrolled distribution of the selected tasks. The Test Director and the Test Conductor must agree on a reasonable time limit for distribution of the maintenance task list to allow adequate time for coordination of engineering and technical support. Releasing the list 2 weeks prior to the demonstration should be adequate if the candidate faults are properly validated.

f. The test center representative is responsible to witness the test, advise the Test Director, and provide assistance in identifying maintainability issues. Procedural issues will be discussed with the Test Conductor and corrected through mutual agreement and guidance from the Test Director.

g. Maintenance tasks which are not demonstrated during the PT/LD may be incorporated into the developmental test maintenance evaluation procedures as Dedicated Maintenance Evaluation (DME) tasks if the resources and schedule permit.

h. Witnesses must actively participate in the data authentication process by keeping times, historical notes, and recording test observations. Official times are the responsibility of the designated Timekeeper/Data Recorder. The test witness times are utilized as a tool to verify the time recorded by the Test Team. Any timekeeping discrepancies should be discussed with the Test Director. Test observations should provide sufficient detail to allow the preparation of TIRs.

i. Operator and maintenance personnel should be debriefed by the Test Conductor at the conclusion of each maintenance task, where problems are apparent. This will reduce confusion and allow for more accurate data collection since it becomes more difficult to distinguish the individual tasks as the volume of maintenance increases. It is not necessary to sequester operator and maintenance personnel during subsequent discussion of the maintenance tasks with the exception of troubleshooting certification. Discussion must be limited to the Test Conductor and test personnel during active maintenance since outside assistance will influence the test results.

j. Test participants and observers must be afforded the opportunity to address areas of concern. The best mechanism is to provide a central location for blank data sheets and a designated collection box. The Data Recorder should be responsible to collect the sheets and compile the comments.

k. No retests are anticipated in the PT/LD since the majority of demonstration tasks involve mechanical procedures. Any tasks which cannot be performed should be verified at a later date after appropriate engineering review and validation of maintenance procedures. In the event a task is retested, the demonstration task will retain the same unique number assigned during the selection process but will be annotated as "Retest".

l. A post test meeting/debriefing will be conducted at the conclusion of



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

test activities. The test observations should be reviewed and appropriate comments solicited from test observers. Preliminary test findings may be presented if they are available.

m. The same test conduct errors that typically occur in the M-Demo may occur in the PT/LD. However, they may not significantly impact the test results since specification compliance is not the test objective.

5. DATA REQUIRED.

5.1 Maintainability Demonstration (M-Demo).

a. Form B-5, "M-Demo Data Sheet", is provided in Appendix B. This data sheet identifies the specific data requirements for maintenance tasks and accommodates data collection for the standard maintenance categories of Crew (C), Organizational (O), Direct Support (F), General Support (H), and Contractor Logistic Support (CLS) at the corresponding maintenance level. These designations conform to current Army maintenance doctrine.

b. Test data must provide sufficient detail to conduct both statistical analysis and technical analysis. This information forms the decision baseline for improving the maintainability of the test item if it does not meet the accept criteria established by the test plan. The technical details surrounding each maintenance task are significant since final analysis will consider probable causes as well as mean time to repair figures.

c. Mechanical design, electrical design, incorporation of BIT and diagnostic routines, human factors engineering (HFE), training, technical publications, personnel skills, and manpower requirements are all issues which affect the maintainability of electronic systems. Data must be collected to assess the impact of these elements on achieving the system requirement.

d. TIR information must provide all the functional details to identify part numbers, functional groups, item name, system nomenclature, and model number.

5.2 Physical Teardown/Logistic Demonstration (PT/LD).

a. Form B-6, "Physical Teardown Data Sheet" is provided in Appendix B. This data sheet identifies the specific data requirements for maintenance tasks.

b. Test data will focus on technical parameters such as number of fasteners, accessibility, tool requirements, support equipment requirements, technical instructions, and maintenance allocation.

c. Statistical data is relatively limited since the nature of the tasks is primarily mechanical.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

d. Form B-7, "Dedicated Maintenance Evaluation (DME) Data Sheet" is provided in Appendix B. This data sheet identifies the specific data requirements for maintenance and troubleshooting tasks and accommodates data collection for the standard maintenance categories of Crew (C), Organizational (O), Direct Support (F), General Support (H), and Contractor Logistic Support (CLS) at the corresponding maintenance level. These designations conform to current Army maintenance doctrine.

e. Test data include the combined requirements of an M-Demo and PT/LD. This allows specific analysis of the troubleshooting methodology.

## 6. PRESENTATION OF DATA.

### 6.1 General.

a. Data presentation requirements will vary depending on whether the test is being conducted by the prime contractor or the government. A test report is typically required under most contract statements. Government witnessed tests will utilize the Test Incident Reporting System (TIRs) as the primary means of providing test results to the development community. The responsible TECOM test center will review any contractor reports and provide independent comments based on their test observations. This is essential since the government's conclusions may significantly differ from the contractor's.

b. When a government report is required, the appropriate format is a separate letter report with data sheets included as appendices. Timely preparation and distribution of the report is crucial to resolution of maintainability problems within the scope of the contract.

c. The least preferred method of data presentation is to include the test results as a subtest in the DT report. Because lag time between the demonstrations and distribution of the DT report typically exceeds 6 months, there is little remaining opportunity to invoke any corrective action.

d. The report will contain a summary and narrative evaluation of the data presented on the forms. Classify and identify all deficiencies and shortcomings, make recommendations as appropriate, and provide statistical calculations to support an accept/reject decision. DOD Handbook 791[8] provides maintainability design techniques.

### 6.2 Maintainability Demonstration (M-Demo).

a. A test start TIR is prepared and submitted to document the pretest briefing, personnel comprising the test team, administrative details, and the start of the first maintenance task activities.

b. Prepare TIRs for each maintenance task demonstrated or attempted during the test. This provides a historical record which is included in the



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

Test and Evaluation Master Database.

(1) The incident description will provide the following header information:

MAINTAINABILITY DEMONSTRATION
TASK No.:
CANDIDATE FAULT No.:
FAULTY COMPONENT:
METHOD OF SIMULATION:
OPERATIONAL SYMPTOM:

(2) The supporting narrative will recreate the events of the demonstration, identify test observations, and provide the measured maintenance value in hours and minutes. A determination statement will be included to identify any problems detected during performance of the maintenance task.

(3) M-Demo tasks are considered Essential Maintenance Actions (EMA) for M-Demo TIR purposes under the RAM category.

(4) Individual task chargeability is normally determined under the following guidelines: Information - No problems; Minor - Some problems but task was completed; Major - Task not completed or aborted for other than administrative reasons.

(5) Maintenance supportability problems which require independent resolution are reported by separate TIR. Specific details are required in the narrative. General impact statements will be included in the individual task TIR. All associated data will be correlated by cross referencing TIR numbers.

c. A summary TIR is prepared to provide concluding administrative remarks, report the end of test, consolidate the maintenance task times, and report the overall test results.

(1) The incident description will provide the following consolidated/task data:

| TASK No. | CANDIDATE No. | M TIME (HRS:MIN) | STATUS | REMARKS |
|---|---|---|---|---|

(2) The appropriate formula provided in MIL-STD-471A[1] will be utilized to calculate the test results. They will be presented as calculated test results and associated with the criteria for the accept/reject decision.

(3) Overall demonstration chargeability is determined as Information if the system meets the accept criteria and Major if the system meets the reject criteria. A chargeability of Minor may be appropriate when the test results are within 10 percent of the reject criteria.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

6.3 Physical Teardown/Logistic Demonstration (PT/LD).

a. The same general data requirements apply to the PT/LD. However, since the data is not used to calculate an accept/reject decision no presentation methodology is established.

b. A test start TIR is prepared and submitted to document the pretest briefing, personnel comprising the test team, administrative details, and the start of the first maintenance task activities.

c. Prepare a summary TIR for all physical teardown maintenance tasks. The incident description will utilize a top down structure by functional group code (FGC) and provide the following task data:

| TASK No. | FGC | M FUNCTION | ITEM NAME | M TIME (HRS:MIN) | MAC TIME |
|---|---|---|---|---|---|

d. Individual TIRs are required only for those tasks which had significant problems. The header data will be provided as follows:

PHYSICAL TEARDOWN

TASK No.:
FGC:
M FUNCTION:
ITEM NAME:
MAC TIME:
PROBLEM AREAS:

e. Individual TIRs are required only for those tasks which encounter significant problems. The summary TIR will provide historical data on ALL tasks performed during the test. The same data requirements identified for the M-Demo also apply here plus some additional header information.

PT/LD TROUBLESHOOTING

TASK No.:
CANDIDATE FAULT No.:
FAULTY COMPONENT:
METHOD OF SIMULATION:
OPERATIONAL SYMPTOM:
FLOW CHART/PROCEDURE NAME:

f. A final TIR will be prepared to provide concluding administrative remarks, report the end of test, and report the overall test results.

(1) The incident description will provide a summary of test observations and subjective comments regarding their impact on system maintainability. Where applicable, data will be quantified and supporting technical



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

details will be provided.

(2) Quantified data will describe the proportion and number of maintenance tasks which deviate from the predicted MAC value; tasks with tool, publication, training, skills, or manpower shortcomings; tasks where the troubleshooting procedures were incorrect; accuracy of BIT and diagnostic routines; human factors deficiencies; and design issues.

(3) Individual task chargeability is normally determined under the following guidelines: Information - No problems; Minor - Some problems but task was completed; Major - Task not completed or aborted for other than administrative reasons.

(4) Test results will be further categorized by applying a weighted scale to the deficiencies and shortcomings recorded during the test. A quality index (QI) will be calculated in accordance with the rules provided in Appendix C. An QI of 70 or greater is considered acceptable.

6.4 Data Forms.

Retain a completed set of the original handwritten data forms in project files to support conclusions set forth in test reports. Be sure the forms are legible and authenticated with the required signatures. Data recording errors are to be corrected with a single pen stroke and initialed by the data recorder. If the data forms must be typed for legibility, all comments on the original form will be transcribed to the typed copy. Original signatures are required on all forms.

Recommended changes to this puiblication should be forwarded to Commander, US Army Test and Evaluation Command, ATTN: AMSTE-CT-T, Aberdeen Proving Ground, MD 21005-5055. Technical information may be obtained from the preparing activity: Commander, US Army Electronic Proving Ground, ATTN: STEEP-MO, Fort Huachuca, AZ 85613-7110. Additional copies are available from the Defense Technical Information Center, Cameron Station, Alexandria, VA 22304-6145. This document is identified by the accession number (AD No.) printed on the first page.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

# APPENDIX A. BACKGROUND

Materiel readiness is a key measure of a military force's ability to enter into and sustain a defense posture. Unit readiness goals are established after careful review of the using unit mission profile, number of items supporting the mission, and existing resources. Achievement of these goals is highly dependent on the user's ability to return failed items to an operational status as quickly as possible.

Since maintainability characteristics are inherent to the design of an item, product development must utilize an integrated approach. Concurrent Engineering ensures equipment design, maintenance planning, diagnostic capabilities, support equipment, training, and personnel requirements provide the optimum union of performance, life cycle costs, and support considerations. Army Regulation 702-3[8], paras 1-8 and 2-10 identifies RAM considerations.

Maintainability is a measure of an item's ability to be retained in or restored to a specified condition within a specific time limit. Measurement of these characteristics provides essential data to management for reduction of life cycle costs and manpower requirements, and improvement of operational readiness. Army Regulation 73-1[9], para 4-2 defines pre-milestone III testing and logistic demonstration.

The primary mechanism for qualifying materiel systems is through the Maintainability Demonstration and the Physical Teardown/Logistic Demonstration. These tests are typically conducted as part of the prime contract and provide verification that the item complies with the specifications. Army Regulation 700-127[2], para 3-27 identifies logistic issues.

Integrated Logistic Support (ILS) testing provides additional maintenance evaluation data based on actual reliability failures while the equipment is operating under a representative mission profile. Army Regulation 700-127[2], para 3-28 identifies supportability issues.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## APPENDIX B. DATA COLLECTION FORMS

### INVENTORY/INSPECTION DATA SHEET

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

| COMPONENT INVENTORY LIST | | |
|---|---|---|
| Line Replaceable Units | Serial Number | COEI Number |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

| SOFTWARE INVENTORY LIST | | |
|---|---|---|
| Operational Software Title | Version Number | Serial Number |
| | | |
| Diagnostic Software Title | Version Number | Serial Number |
| | | |

Form B-1



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## INVENTORY/INSPECTION DATA SHEET (Cont)

Test Title: _____ Date: _____

Nonmenclature: _____

Model No.: _____ Serial No.: _____

SUPPORT ITEM INVENTORY LIST

| Support Item Name | Serial Number | Calib/Rmks |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

COMMENTS

_____
_____
_____
_____
_____

Test Conductor: _____ Date: _____
Test Witness: _____ Date: _____
Test Director: _____ Date: _____

Form B-1



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## CONFIGURATION DATA SHEET

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

Configuration Description: _____

TEST SET-UP ILLUSTRATION

Test Conductor: _____ Date: _____
Test Witness: _____ Date: _____
Test Director: _____ Date: _____

Form B-2



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## SYSTEM STATUS DATA SHEET

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

### BUILT IN TEST RESULTS

1. System Power Up: _____

2. Manual BIT: _____

3. Diagnostics: _____

### TECHNICAL MANUAL REFERENCES

### CORRECTIVE ACTION DESCRIPTION

Test Conductor: _____ Date: _____
Test Witness: _____ Date: _____
Test Director: _____ Date: _____

Form B-3



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

# TEST TEAM DATA SHEET

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

---

## TEST TEAM

1. Test Director: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____
   _____
   _____
   _____

2. Test Conductor: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____
   _____
   _____
   _____

3. Test Witness: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____
   _____
   _____
   _____
   _____

4. Fault Technician: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____
   _____
   _____
   _____
   _____

Form B-4



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## TEST TEAM DATA SHEET (Cont)

Test Title: _____ Date: _____

TEST TEAM (CONTINUED)

5. System Operator: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____

6. Maintainer: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____

7. Timekeeper: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____

8. Data Recorder: _____
   Organization: _____

   Phone (DSN/Comm): _____
   Mailing Address: _____

Form B-4



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

**TEST TEAM DATA SHEET (Cont)**

Test Title: _____ Date: _____

TEST TEAM (CONTINUED)

9. _____ : _____
Organization: _____

Phone (DSN/Comm): _____
Mailing Address: _____

10. _____ : _____
Organization: _____

Phone (DSN/Comm): _____
Mailing Address: _____

11. _____ : _____
Organization: _____

Phone (DSN/Comm): _____
Mailing Address: _____

12. _____ : _____
Organization: _____

Phone (DSN/Comm): _____
Mailing Address: _____

Form B-4



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## MAINTAINABILITY DEMONSTRATION DATA SHEET

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

---

Date: _____ Start Time: _____ End Time: _____
Maintenance Category: __ C __ O __ F __ H __ CLS(O/F/H/D))
Sequence _____ Task Number _____ Candidate Fault Number _____
Operator: _____ Maintainer: _____
Status: __ Complete __ Aborted __ Retest __ Invalid (Delete)

---

Fault Insertion: Complete: _____ Symptom Verified: __ yes __ no
Detection: _____ Method: __ BIT __ Operation Program __ Manual
Symptom Briefed: _____

_____

_____

TM Procedure Followed: _____

_____

---

TASK DATA

Isolate: _____ _____ _____ _____ Total: _____

Remove: _____ _____ _____ _____ Total: _____

Install: _____ _____ _____ _____ Total: _____

Verify: _____ _____ _____ _____ Total: _____

Task Total: _____

---

COMMENTS/OBSERVATIONS

_____

_____

_____

_____

_____

_____

Test Conductor: _____ Date: _____
Test Witness: _____ Date: _____
Test Director: _____ Date: _____

Form B-5



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

**PHYSICAL TEARDOWN DATA SHEET**

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

Date: _____ Start Time: _____ End Time: _____
Sequence _____ Task Number _____ Candidate Task Number _____
Operator: _____ Maintainer: _____
Status: __ Complete __ Aborted __ Retest __ Invalid (Delete)

| TASK DATA | | | |
|---|---|---|---|
| FGC | COMPONENT/ASSEMBLY | | PART NUMBER |
| | | | |
| FUNCTION | MAC TIME | LEVEL | NEXT HIGHER ASSEMBLY |
| | | | |

| MAINTENANCE TIMES | | | |
|---|---|---|---|
| Inspect | | Remarks | |
| Disassemble | | Remarks | |
| Remove | | Remarks | |
| Install | | Remarks | |
| Assemble | | Remarks | |
| Test | | Remarks | |
| Inspect | | Remarks | |

| SUPPORT REQUIREMENTS |
|---|
| Publication Reference |
| Tools Required |

Form B-6



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## PHYSICAL TEARDOWN DATA SHEET (Cont)

Test Title: _____ Date: _____

### SUPPORT REQUIREMENTS (Continued)

TMDE Required _____

Personnel Required _____

Safety _____

### ISSUES

__ Maint. Planning __ Design __ Publications __ Tools __ TMDE
__ Safety __ HFE/MANPRINT __ Training __ Personnel __ Manpower

Test Conductor: _____ Date: _____
Test Witness: _____ Date: _____
Test Director: _____ Date: _____

Form B-6



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## DEDICATED MAINTENANCE EVALUATION (DME) DATA SHEET

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

---

Date: _____ Start Time: _____ End Time: _____
Maintenance Category: __ C __ O __ F __ H __ CLS(O/F/H/D)
Sequence ____ Task Number _____ Candidate Fault Number _____
Operator: _____ Maintainer: _____
Status: __ Complete __ Aborted __ Retest __ Invalid (Delete)

---

### TROUBLESHOOTING DATA

Simulated Failure: _____
Method of Simulation: _____
_____
Diagnostic Results: _____
_____

---

| FUNCTION | START | STOP | TIME | REMARKS |
|---|---|---|---|---|
| Detect | ___:___ | ___:___ | _____ | _____ |
| Isolate | ___:___ | ___:___ | _____ | _____ |
| Repair | ___:___ | ___:___ | _____ | _____ |
| Verify | ___:___ | ___:___ | _____ | _____ |

TOTAL TASK TIME: ___.___ DECIMAL _____ MAC

---

### MAINTENANCE SUPPORT DATA

Initial BIT Indication: _____
_____
Subsequent Indications: _____
_____
_____
Diagnostic Indication: _____
_____
Ambiguity Group: _____
_____

Form B-7



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## DME DATA SHEET (Cont)

Test Title: _____ Date: _____

MAINTENANCE SUPPORT DATA (Continued)

Publication: _____

Tools Required: _____

Tools Used: _____

TMDE Required: _____

TMDE Used: _____

Manpower Required: _____

Manpower Used: _____

ISSUES

__ Maint. Planning __ Design __ Publications __ Tools __ TMDE
__ Safety __ HFE/MANPRINT __ Training __ Personnel __ Manpower

Test Conductor: _____ Date: _____
Test Witness: _____ Date: _____
Test Director: _____ Date: _____

Form B-7



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

# APPENDIX C. QUALITY INDEX RULES AND CRITERIA

## FOR LOGISTIC/MAINTAINABILITY ANALYSIS

This process establishes a combined quantitative/subjective evaluation guideline for analysis of logistic support maintenance characteristics.

Assessment Threshold. A value of ten (10) assessment points is awarded for each maintenance task demonstrated. This value is established as the 100th percentile evaluation criteria. If thirty tasks are demonstrated the maximum allowable assessment points is 300.

Quantitative Deductions. The following are subtractive point values for technical deficiencies and shortcomings. Each incident classification is awarded points. Additional points are awarded based on the problem impact. A MAJOR incident is awarded a point value of minus ten (-10). If the problem impact is a publication error which must be corrected prior to DT/OT an additional point value of minus five (-5) is awarded.

INCIDENT CLASSIFICATION

| | |
|---|---|
| MAJOR | 10 Points |
| MINOR | 3 Points |

PROBLEM IMPACT
(Fix Before Fielding/Fix Before DT/OT)

| | |
|---|---|
| TOOL | 2 Points/5 Points |
| TRAINING | 2 Points/5 Points |
| SKILL | 2 Points/5 Points |
| MANPOWER | 2 Points/5 Points |
| PUBLICATION | 2 Points/5 Points |
| DESIGN | 2 Points/5 Points |
| SAFETY | 2 Points/5 Points |

Quality Deductions. The following are subtractive point values for overall quality deficiencies recorded during the test. A weighted scale is established based on the probable impact to system supportability. Test results which document 50 percent or greater error in predicted MAC times are awarded a point value of minus three (-3)

Analysis Criteria. The following Quality Index Scale is established as assessment criteria:

QI ASSESSMENT CRITERIA

| | |
|---|---|
| 100 - 90 percentile QI | A |
| 90 - 80 percentile QI | B |
| 80 - 70 percentile QI | C |
| 70 - 0 percentile QI | Reject |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

## QUALITY INDEX DATA SHEET

Test Title: _____ Date: _____

Nomenclature: _____

Model No.: _____ Serial No.: _____

### ASSESSMENT THRESHOLD

| | | Eval Criteria |
|---|---|---|
| ____ Tasks X 10 Value Points = | = _____ | _____ |

### QUANTITATIVE DEDUCTIONS

____ Major Incidents X 10 Value Points = _____
____ Minor Incidents X 03 Value Points = _____

____ Tool Problems X 02 Value Points = _____
____ Trng Problems X 02 Value Points = _____
____ Skill Problems X 02 Value Points = _____
____ Manpwr Problems X 02 Value Points = _____
____ Pubs Problems X 02 Value Points = _____
____ Design Problems X 02 Value Points = _____
____ Safety Problems X 02 Value Points = _____

____ Tool Problems X 05 Value Points = _____
____ Trng Problems X 05 Value Points = _____
____ Skill Problems X 05 Value Points = _____
____ Manpwr Problems X 05 Value Points = _____
____ Pubs Problems X 05 Value Points = _____
____ Design Problems X 05 Value Points = _____
____ Safety Problems X 05 Value Points = _____

____ Total Quantitative Deductions = _____

### QUALITATIVE DEDUCTIONS

____ MAC times in error > 50% = 3 Points
____ Tasks w/TM errors > 30% = 3 Points
____ Troubleshooting errors > 20% = 3 Points
____ Tasks w/HFE problems > 10% = 3 Points
____ Bit Isolation errors > 5% = 3 Points

| | Deductions | QI |
|---|---|---|
| ____ Total Qualitative Deductions = | _____ | _____ |

Quality Index (Eval Criteria minus Deductions) _____
Accept Decision = QI > 70% Eval Criteria

Form C-1



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

# APPENDIX D. REFERENCES

## REQUIRED DOCUMENTATION

1. MIL-STD-471A, Maintainability Verification/Demonstration/ Evaluation, dated 8 December 1988.

2. Army Regulation 700-127, Integrated Logistic Support, dated 17 July 1990.

3. Army Regulation 200-2, Environmental Effects of Army Actions, dated 23 December 1988.

4. Army Regulation 750-1, Army Materiel Maintenance Policy and Retail Maintenance Operations, dated 30 October 1989.

5. MIL-STD-721C, Definition of Terms For Reliability and Maintainability, 23 October 1991.

6. MIL-STD-1388-1, Logistic Support Analysis, dated 28 March 1991.

7. MIL-STD-1388-2, DoD Requirements for a Logistic Support Analysis Record, dated 28 March 1991.

8. Army Regulation 702-3, Army Materiel Systems, Reliability, Availability, and Maintainability, dated 31 March 1987.

9. Army Regulation 73-1, Test and Evaluation Policy, dated 15 August 1992.

## INFORMATIONAL DOCUMENTATION

a. DA Pamphlet 700-28, ILS Program Assessment and Criteria, dated 4 December 1987.

b. DA Pamphlet 738-750, Functional Users Manual for The Army Maintenance Management System (TAMMS), Update 12 dated 31 October 1989.

c. MIL-STD-470B, Maintainability Program for Systems and Equipment, dated 30 May 1989.

d. MIL-STD-472, Maintainability Prediction, dated 12 January 1984.

e. MIL-STD-280A, Definition of Item Levels, Item Exchangeability, Models, and Reliability Terms, dated 30 July 1986.

f. MIL-STD-1472D, Human Engineering Design Criteria for Military Systems, Equipment and Facilities, dated 30 June 1992.

g. MIL-STD-1629A, Procedures for Performing a Failure Mode Effects and Criticality Analysis, dated 28 November 1984.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:51Z
Check the source to verify that this is the current version before use.